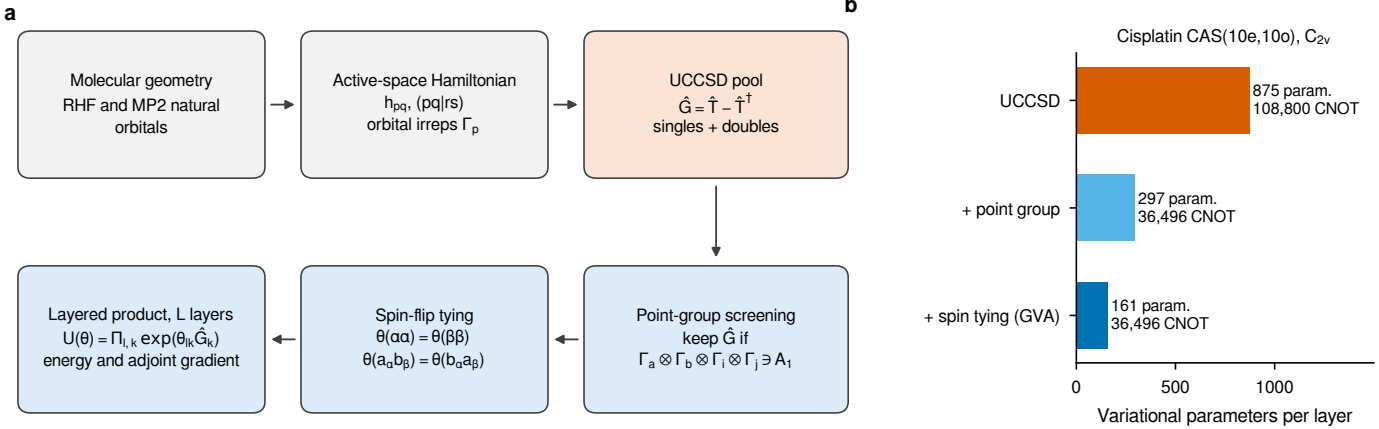
a
Molecular geometry
RHF and MP2 natural orbitals
Active-space Hamiltonian
h_{pq}, (pq|rs)
orbital irreps Γ_p
UCCSD pool
$\hat{G} = \hat{T} - \hat{T}^\dagger$
singles + doubles
Point-group screening
keep $\hat{G}$ if
$\Gamma_a \otimes \Gamma_b \otimes \Gamma_i \otimes \Gamma_j \ni A_1$
Spin-flip tying
$\theta(\alpha\alpha) = \theta(\beta\beta)$
$\theta(a_\alpha b_\beta) = \theta(b_\alpha a_\beta)$
Layered product, L layers
$U(\theta) = \Pi_{l,k} \exp(\theta_{lk} \hat{G}_k)$
energy and adjoint gradient
b
Cisplatin CAS(10e,10o), C_{2v}
UCCSD
875 param.
108,800 CNOT
+ point group
297 param.
36,496 CNOT
+ spin tying (GVA)
161 param.
36,496 CNOT
0
500
1000
Variational parameters per layer